

EE 115 Homework 6

For demodulation of an FM signal $u_{FM}(t)$, one method involves a differentiator. Let the input of an ideal differentiator be $u_{FM}(t) = A_c \cos(2\pi f_c t + \theta(t))$ with $\theta(t) = 2\pi k_f \int_0^t m(\tau) d\tau$. Then the output of the ideal differentiator is

$$y(t) = \frac{d}{dt} u_{FM}(t) = -2\pi A_c (f_c + k_f m(t)) \sin(2\pi f_c t + \theta(t)) \quad (1)$$

from which we can use an envelope detector and an DC blocker to retrieve the message $m(t)$.

- 1) **(50 points)** To implement an differentiator, one can try a RL circuit whose transfer function is

$$H_{RL}(s) = \frac{sL}{R + sL}. \quad (2)$$

Let the bandwidth of $u_{FM}(t)$ be W , what is a proper condition on R , L , f_c and W such that the RL circuit can be used in place of the differentiator? (Hint: What is the highest frequency in $u_{FM}(t)$ and how does it relate to the answer?)

- 2) **(50 points)** Another method to implement a differentiator is an RLC circuit whose transfer function is

$$H_{RLC}(s) = \frac{\frac{1}{sC + \frac{1}{sL}}}{R + \frac{1}{sC + \frac{1}{sL}}} = \frac{1}{RC} \cdot \frac{s}{s^2 + \frac{1}{RC}s + \frac{1}{LC}}. \quad (3)$$

The poles of this function are $p_{1,2} = \frac{1}{2} \left(-\frac{1}{RC} \pm \sqrt{\left(\frac{1}{RC}\right)^2 - \frac{4}{LC}} \right)$. If $\left(\frac{1}{RC}\right)^2 \ll \frac{4}{LC}$, then

$$p_{1,2} = -\sigma \pm j\omega_0 \quad (4)$$

with $\sigma = \frac{1}{2RC}$ and $\omega_0 = 2\pi f_0 = \frac{1}{\sqrt{LC}}$ (which is the resonant angular frequency of the circuit).

- a) Sketch $|H_{RLC}(j2\pi f)|$ to illustrate that there is a value f_c (between 0 and f_0) around which $|H_{RLC}(j2\pi f)|$ can be approximated linearly as

$$|H_{RLC}(j2\pi f)| \approx H_0 + \alpha(f - f_c) = c + \alpha f \quad (5)$$

where $c = H_0 - \alpha f_c$. Provided that the above approximation holds well for $|f - f_c| < W/2$, the RLC circuit can also be used in place of the differentiator.

- b) What are the advantages of the RLC circuit over the RL circuit? (Hint: If $u_{FM}(t)$ contains additive broadband noise, which of the two circuits can filter out more noise? Which of them has a higher gain as a differentiator?)